

Yorkshire Energy Park - Energy Statement

INTRODUCTION

This Energy Statement has been prepared by Vic Coupland Ltd. to support a Reserved Matters Application (RMA) at the Yorkshire Energy Park (YEP). The YEP site is a mixed use development which benefits from an Outline Planning Permission (OPP) (Ref. 17/01673/STOUTE), approved 22 December 2020 by East Riding of Yorkshire Council (ERYC). The Site is located at Land North West Of Kingstown Hotel, Hull Road, Hedon, East Riding Of Yorkshire.

The applicant (Hull Eco Park Ltd (HEPL)) has submitted a RMA for the following proposed development:

Reserved matters application for layout, scale, appearance and landscaping for the construction of an energy centre, a data centre, and associated roads and infrastructure, to outline planning permission 17/01673/STOUTE

Condition 77 on the OPP states the following:

The details submitted pursuant to condition 2 for each part of the development within the Business Park Land Use Zone or the Data Centre Land Use Zone (shaded yellow and purple respectively on plan reference YEP_BDP(PL)003 Rev E), shall include an energy statement demonstrating that the proposed occupier would have specific energy use requirements which would be met by the on-site energy generation facilities. Each part of the development shall be occupied in accordance with the details set out in the approved energy statement. A new or updated energy statement shall be submitted to and approved in writing by the Local Planning Authority before any part of the development within the Business Park Land Use Zone or the Data Centre Land Use Zone is occupied by a subsequent occupier. Subsequent occupation shall thereafter be in accordance with the details set out in the approved new or updated energy statement.

The Data Centre element of this RMA is located within the Data Centre Land Use Zone (as per the approved Land Use Parameter Plan (Ref. YEP_BDP(PL)003 Rev E). Therefore, in line with the requirements of Condition 77, an Energy Statement is required to be submitted alongside this RMA. The Energy Statement is required to demonstrate that the proposed occupier of the Data Centre would have specific energy use requirement which would be met by the on-site energy generation facilities. On-site energy generation will be created by the Energy Centre which also forms part of this RMA. Condition 77 also requires each part of the development (i.e. the Data Centre that forms part of this RMA) to be occupied in accordance with the details set out within the approved Energy Statement.

This Energy Statement, prepared by Vic Coupland Ltd., provides the detail required in order to satisfy the requirements of Condition 77 on the OPP.

ENERGY STATEMENT

The Data Centre operator have specifically selected the YEP site for their new data centre due to the resilient on-site zero carbon electrical and cooling energy generation planned for 2026, when the hydrogen source becomes available, where energy costs are provided at market rates equivalent to the grid supply. The Combined Heat and Power (CHP) turbines will initially be running off natural gas until the planned hydrogen supply becomes available.

The Data Centre is designed to be initially built for 240 data racks requiring approximately 3-4MW of electrical demand with future expansion to 600 racks expanding to a future electrical demand of 6MW. The cooling provided as a by-product of the energy generation, piped in cold water pipes from the energy centre to the data centre, eliminating the need for their own chiller plant.

As the first tenants on the site, the energy centre generation capacity has been designed to align with the requirements of the data centre for both their energy and resilience

Therefore, energy, for the Data Centre, will be drawn from the onsite energy centre (both of which form part of this RMA) which will provide the electrical and cooling requirements of the Data Centre space and ancillary office space with the Data Centre. The Energy Centre will be working towards a zero-carbon footprint eventually adopting hydrogen as its source of energy supply in the future.

Electrical energy provided by a CHP plant will be utilised to provide power to energy efficient technologies were applicable/available.

Cooling energy will be provided through the use of a community system via the proposed energy centre as a by-product of the electrical energy production via the CHP plant.

Heating energy will be provided using a combination of Air Source Heat Pumps and Variable Refrigerant Flow (VRF) systems.

Passive energy efficiency design measures invariably prove to be the most economical and socially acceptable inclusions on any scheme. Ensuring air-tight, well insulated buildings with good control measures are among the most important considerations.

The design team will carry out relevant calculations and modelling to ensure these elements are optimised wherever possible within the scope of the project.

The following provides a summary of the energy efficiency measures that are to be considered in order to improve the feasible energy rating for the new development:

1. The use of glazing to maximise daylight in all areas.
2. Passive solar control to help minimise the requirement for mechanical cooling by use of; solar controlled glazing, deep window reveals and recessed facades to South facing elevations.
3. Partial or full solar control glazing to the south facing glazing as a minimum.
4. Enhanced building thermal performance that exceeds building regulation minimum standards.
5. Maximise efficiency of plant and systems.
6. Optimise control and flexibility of the installations.
7. Air-tight construction. Proposed air permeability rate to surpass current regulatory standards.
8. Heating, ventilation, and air conditioning (HVAC) and lighting systems to operate 'on-demand' wherever practical.